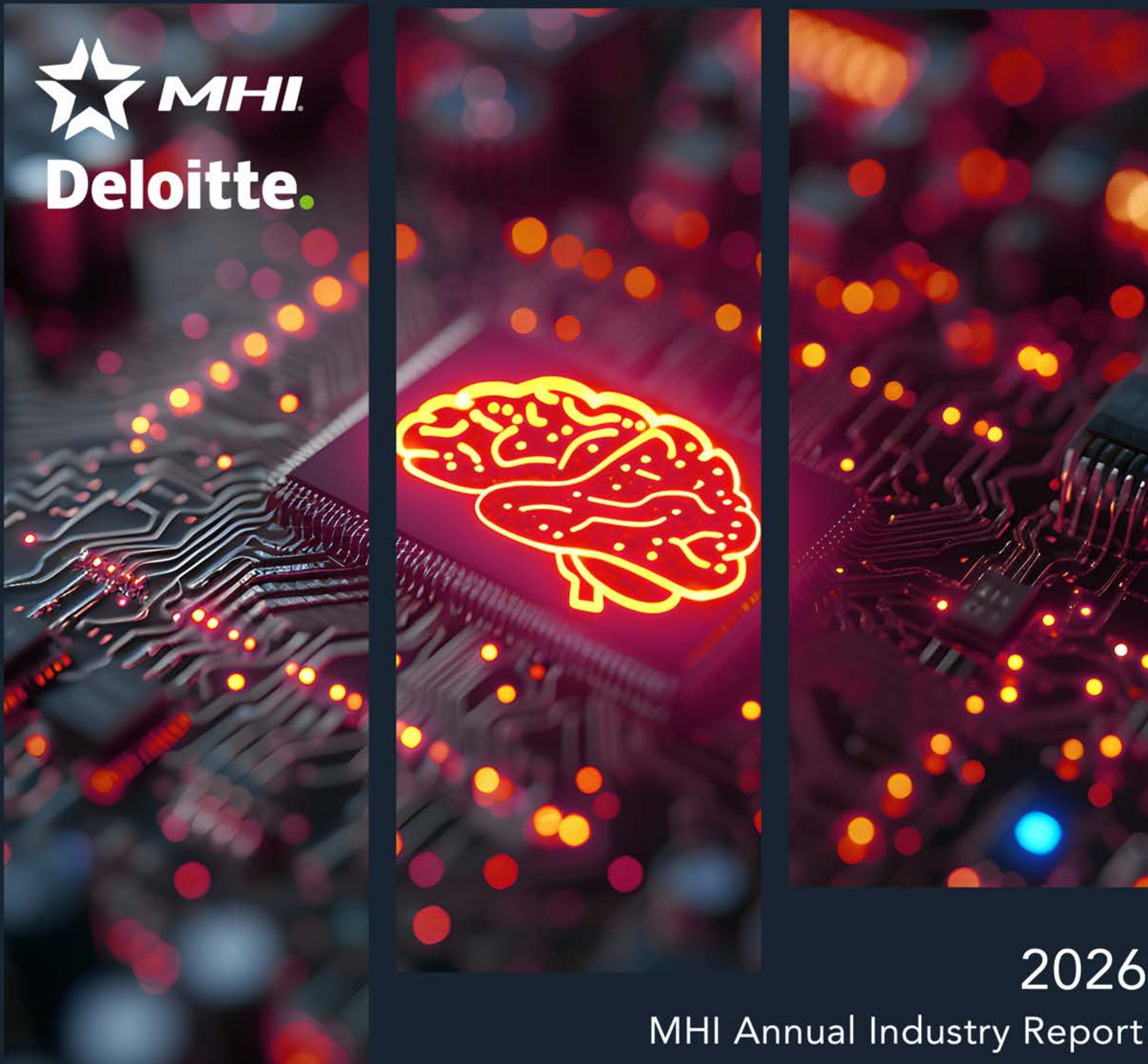

2026

MHI Annual Industry Report

REWIRING THE FUTURE

A Supply Chain Playbook for Innovation

REWIRING THE FUTURE

A SUPPLY CHAIN PLAYBOOK FOR INNOVATION

2026 MHI Annual Industry Report Key Findings

ADOPTION TRENDS

Predicted adoption of technologies within 5 years

In-use today / Within 5 years

	In-use today	Within 5 years
Artificial Intelligence	41%	47%
Advanced Analytics	38%	48%
Cloud Computing	54%	31%
Internet of Things/Sensors	40%	37%
Robotics & Automation	35%	38%
Wearable & Mobile Technology	28%	41%
Autonomous Vehicles & Drones	16%	34%
Humanoid Robotics	4%	28%

TOP SUPPLY CHAIN CHALLENGES

Top 5 company challenges rated as a major challenge

58%	Talent acquisition & workforce challenges
48%	Accurate forecasting & inventory management
46%	Meeting evolving customer demands
42%	Managing cyber risks & competitive pressures
42%	Supply chain synchronization & visibility

TOP SUPPLY CHAIN TRENDS

Ranked trends based on impact to operations

1. Economic uncertainty & inflation
2. Workforce & talent shortage
3. Pace of technology adoption
4. Supply chain visibility, agility, & resiliency
5. Cybersecurity and data security

ADOPTION BARRIERS

Top 3 company barriers to adopting technologies

CYBERSECURITY CONCERNS	40%
LACK OF CLEAR BUSINESS CASE	36%
LACK OF BUDGET	30%

IMPACT OF TECHNOLOGIES

Potential to disrupt the industry

Artificial Intelligence	70%
Robotics & Automation	64%
Autonomous Vehicles & Drones	53%
Advanced Analytics	53%
Internet of Things/Sensors	47%
Cloud Computing	47%
Humanoid Robotics	45%
Wearable & Mobile Technology	43%

AI USE IN SUPPLY CHAIN

Current or predicted uses of AI within 2 years

33%	Enhancing demand/inventory optimization
30%	Improving predictive maintenance/equipment reliability
27%	Automating decision-making in operations
26%	Optimizing logistics/transportation routes
24%	Streamlining supplier and procurement processes
23%	Detecting and mitigating risk in real time
23%	Personalizing customer experiences/order fulfillment
20%	Accelerating design/prototyping with digital twins

28% not leveraging AI technologies

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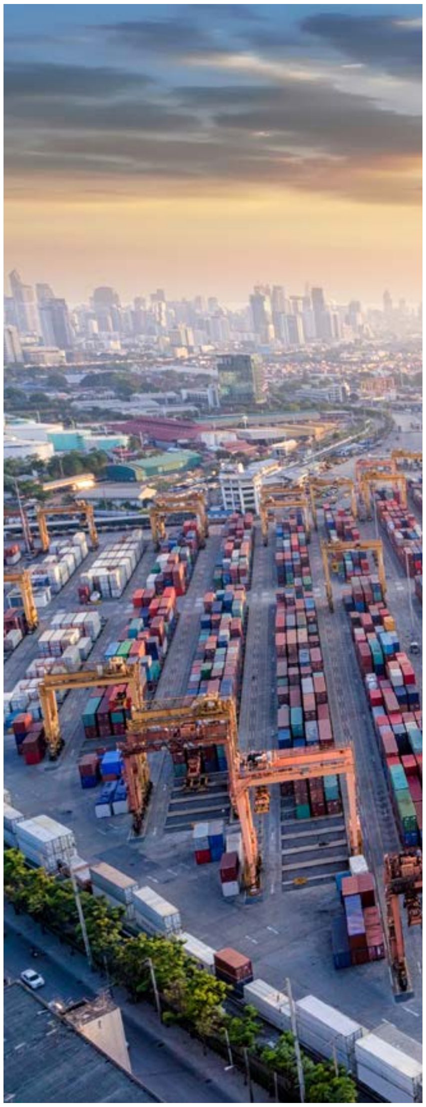


Supply chains can no longer be optimized at the edges—they must be rewired end-to-end. Only connected, intelligent, and automated real-time networks will withstand the volatility and meet the future customer demands for speed and efficiency.

John Paxton, CEO, MHI

INTRODUCTION

Today's supply chain organizations operate in an environment defined by relentless disruption, volatility, and rapidly shifting market demands. Geopolitical uncertainty, labor shortages, accelerating technology cycles, and elevated customer expectations have converged to make predictability a thing of the past. In response, supply chain leaders are reassessing every aspect of their operations, investing not only in advanced digital technologies such as AI, robotics, and real-time analytics to improve efficiency and resilience, but also in their workforces. Strategic priorities now include strengthening risk management, enhancing transparency, enabling rapid fulfillment, and embedding sustainability throughout the supply chain. These investments reflect a broader recognition that future-ready supply chains must be agile, adaptive, and powered by both cutting-edge tools and empowered talent.

Supply chain performance over the next decade will be determined by how effectively leaders translate AI, automation, and analytics into measurable operational outcomes. Competitive advantage will not come from isolated pilots or technology adoption alone, but from disciplined operational assessment, intelligent coordination of decisions, and scalable execution across people, processes, and systems. For supply chain decision makers, the mandate is clear: convert innovation into durable performance gains across cost, service, resilience, and growth.

The 2026 MHI Annual Industry Report provides a practical roadmap for organizations as they navigate this era of transformation. By focusing on the intersection of business and technology, this report goes beyond tech trends to explore how operational assessments, smart automation, data-driven decision making, and new approaches to talent development can be woven together to "rewire" supply chain performance. Through objective survey data, real-world examples, case studies, and actionable frameworks, the report aims to help leaders not just manage disruption but harness it as a driver for continuous innovation and long-term success.

This year's report is based on a global survey of more than 500 supply chain professionals and builds on past years' insights by highlighting the rising importance of creating supply chains that are intelligent, resilient, and AI-orchestrated.

The 2026 findings point to a clear inflection point: performance is increasingly defined not by whether organizations adopt AI and automation, but by how effectively they scale and integrate these technologies with their existing capabilities. The survey results show where pressure is concentrating and where leaders are placing their next bets, so your organization can focus its innovation and technology efforts and investments where they will matter most.

The pages that follow provide clear, actionable insights to help supply chain leaders establish and scale intelligent operations, strengthen resilience, and turn technology investments into sustained performance and strategic advantage.

SURVEY HIGHLIGHTS

This year's survey reveals a number of important supply chain challenges and trends—and provides valuable insight into how organizations are responding.

AI AND ROBOTICS/AUTOMATION ARE EMERGING AS THE BIGGEST DISRUPTORS

Artificial intelligence is viewed as the most disruptive technology for the next decade, with a quarter of respondents (24%) categorizing AI as transformational and nearly half (48%) considering its disruptive impact to be significant or greater—up 25 percentage points since 2025. Robotics & automation follow AI as the second most disruptive technology, with 39% rating its impact as significant or greater, up 16 percentage points (Figure 1).

AI's top ranking suggests that industry professionals recognize the technology's potential to revolutionize decision-making, automate complex workflows, and drive operational efficiencies. Similarly, the rise of robotics & automation reflects a growing reliance on these tools to address persistent labor and productivity challenges. The key takeaway for leaders is that strategic investment in AI and automation should be prioritized, as organizations that proactively adapt and harness these technologies will be better positioned to achieve agility, resilience, and a competitive edge, while those that don't will likely lag behind.

AI BUSINESS CASES AND HIGH AUTOMATION COSTS ARE KEY BARRIERS TO ADOPTION

Despite broad agreement that AI and automation are expected to be highly disruptive and rewarding,

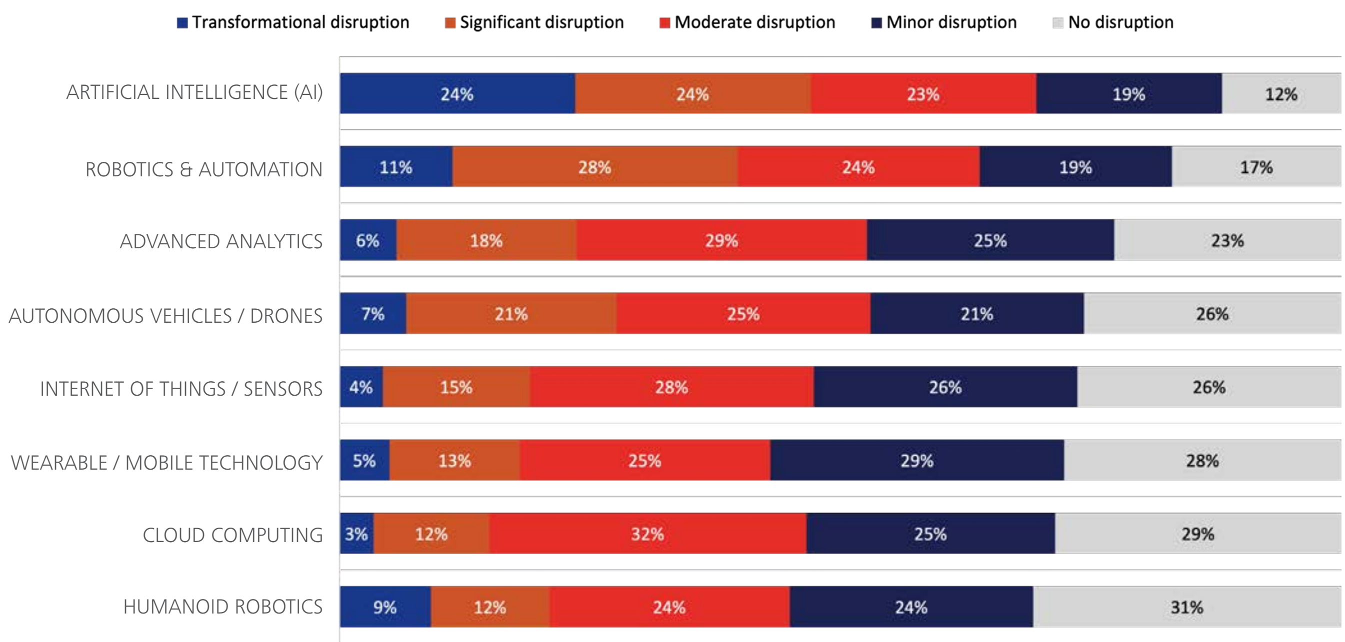


Figure 1: Survey results – disruptive technologies for the next decade

AI - BARRIERS TO ADOPTION

1. building a clear business case
2. lack of understanding
3. lack of adequate talent

Figure 2: Survey results – primary barriers to technology adoption

organizations report significant barriers to adoption (Figure 2). For AI, the top obstacles are: building a clear business case (20%, up 4 percentage points YoY); lack of understanding (19%, down 4 percentage points YoY), and lack of adequate talent (12%, down 2 percentage points YoY) (Figure 2). For robotics & automation, the top hurdle is budget constraints, cited as the primary barrier by nearly a third of respondents (29%, down 3 percentage points YoY).

Even as industry professionals widely acknowledge the power of AI and automation to revolutionize decision-making and help address persistent labor challenges, the pace of progress continues to be impeded by non-technology challenges such as bridging knowledge gaps, securing investment, and upskilling teams.

To unlock the potential of these technologies, leaders must build buy-in, clarify ROI, and align technology adoption with business strategy. Organizations that are slow to overcome such barriers risk missing out on the full benefits of these transformative innovations.

KEY SUPPLY CHAIN CHALLENGES

Talent and workforce issues are affecting 90% of supply chain organizations. Among those, nearly two-thirds rate the challenge as major, up 9 points from last year. This makes attracting, upskilling, and retaining talent a core part of building resilience—and a key requirement for adopting new technology at speed. Managing cyber risks is the second most common challenge (88%). As organizations implement more data-dependent workflows through AI, advanced analytics, and cloud platforms, cybersecurity rises on the agenda (Figure 3).

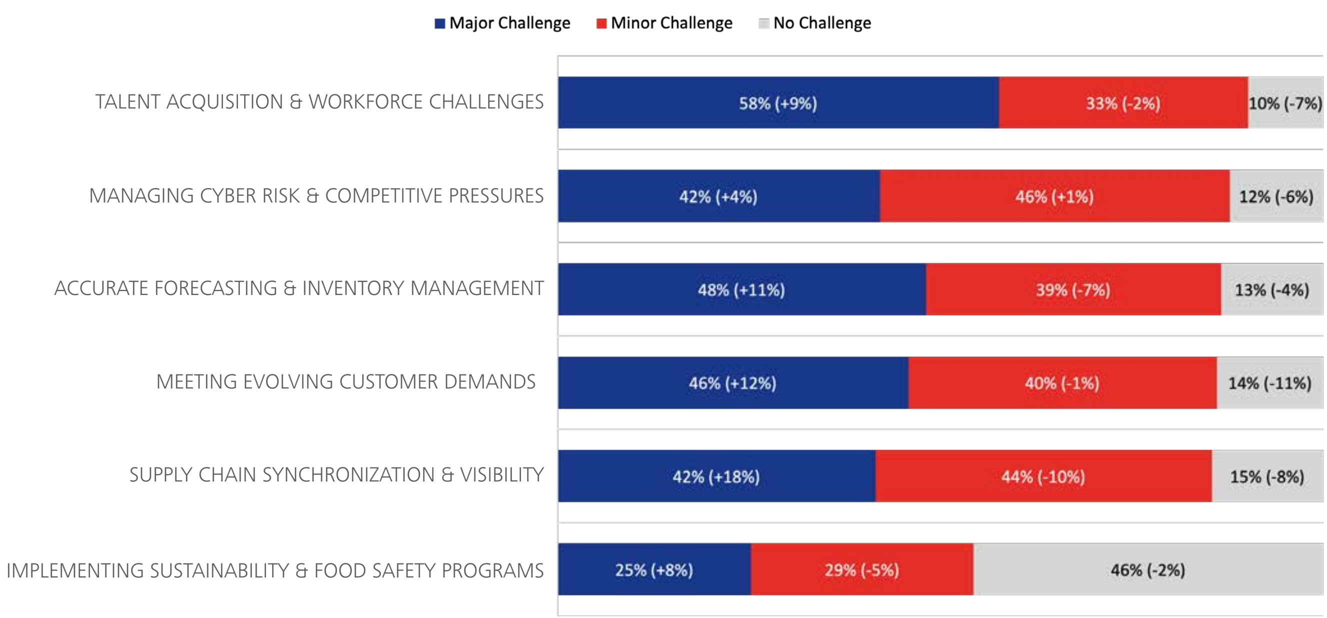


Figure 3: Survey results – organization supply chain challenges

BROAD MARKET TRENDS AFFECTING SUPPLY CHAINS

Economic uncertainty and inflation topped the list of top trends impacting supply chains. Workforce challenges follow closely behind, with pace of technology adoption, supply chain agility/resiliency, and cybersecurity/data security rounding out the top 5 trends affecting supply chains and their leaders.

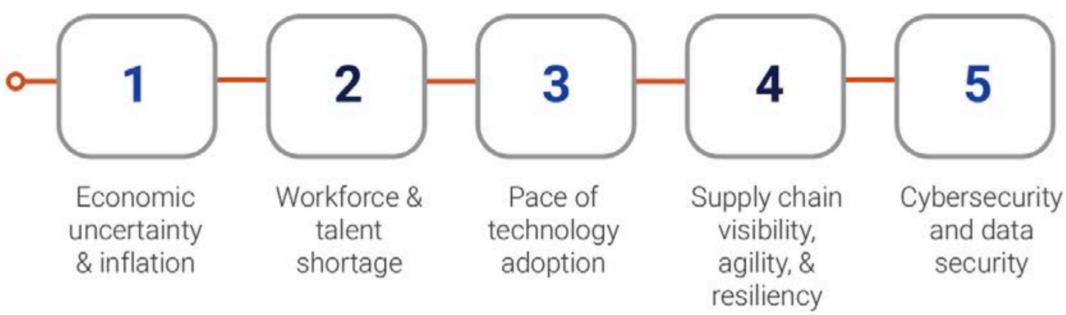
Tariff volatility, interest rate drama, military conflicts, wavering consumer confidence, and ongoing concerns about unemployment are all contributing to a uniquely challenging and unpredictable global economic environment—and are all factors contributing to economic uncertainty and inflation concerns topping the list of top trends impacting supply chains.

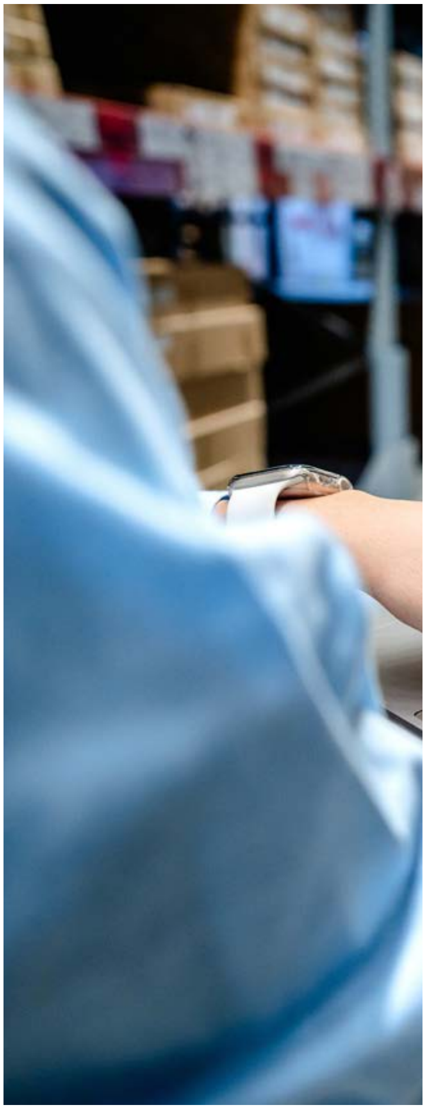
The workforce and talent gap is now the #2 trend affecting supply chains with the the ongoing talent and worker shortage continues as Baby Boomers retire and the adoption of new technologies and automation require reskilling the supply chain workforce.

In the face of these challenges, supply chain leaders are putting more emphasis on technology trends that enable smart operations (AI, real-time data, automation, and emerging technology), which can help them respond quickly when conditions change. Technology trends related to smart operations maintained the #3 and #4 spots in this year's survey. The need for agility, visibility and real-time data in supply chain operations is becoming a higher priority, not just a nice-to have suggesting that running operations with dated equipment and technology is a growing risk when orchestration is necessary for success moving forward.

Cybersecurity, risk & data security moved into the top five. This is likely driven by the growing reliance on cloud systems, AI, and advanced analytics, all of which increase how much companies rely on data (and how much data they store and share).

TOP TRENDS IMPACTING OPERATIONS





AI in decision-making isn't a zero-sum game; it's additive. Its real power is turning massive amounts of data into clear signals that help leaders see what's ahead and act with greater speed and confidence.

Camille Blake, Regional Director, Logistics, Carvana

ELEVATING DECISION MAKING: THE IMPACT OF AI & ANALYTICS

The calculus for supply chain, logistics, and fulfillment leaders has changed. Market volatility and advances in generative AI, agentic AI, physical AI, and edge AI are making real-time supply chain orchestration more attainable. Agility, resiliency, and continuous learning are no longer aspirational capabilities. By combining predictive analytics, machine learning, and edge-based real-time processing with human-in-the-loop (HITL) guardrails, organizations can rewire decision making to be faster and more responsive.

FROM STATIC EXECUTION TO AI-ENABLED ORCHESTRATION

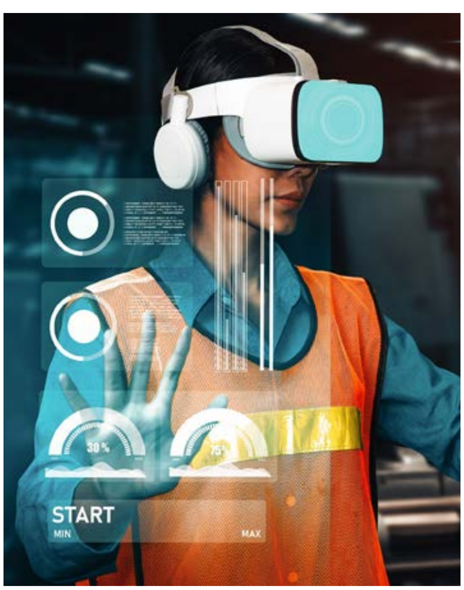
Imagine landing in a new city, picking up a rental car, and relying on a printed static route. Miss a turn or hit standstill traffic and you have no way to adapt. Modern navigation apps close that gap by offering a range of route options with HITL selection, live signals, constantly updated time of arrival, and automatic rerouting. Supply chain and fulfillment are making the same leap from static execution to AI-enabled orchestration. Consider a common fulfillment solution. In a static model, tasks release on a schedule: when a conveyor line or robot goes down, supervisors must scramble to shift labor and/or absorb missed cutoffs. In a dynamic model, exception signals are scored against service-level agreement (SLA) risk, micro-simulations are used to compare options (often via a digital twin), and the system recommends—or executes within designated guardrails—the best next action (with predicted KPI impact), escalating to humans when confidence drops or pre-defined constraints are hit. This represents a foundational shift. The goal is an orchestration layer that connects signals, decisions, execution, and learning so operations can respond as one flexible, adaptable system. Done well, it reduces the coordination tax that shows up when people act as the "human API," manually bridging disconnected tools and teams.

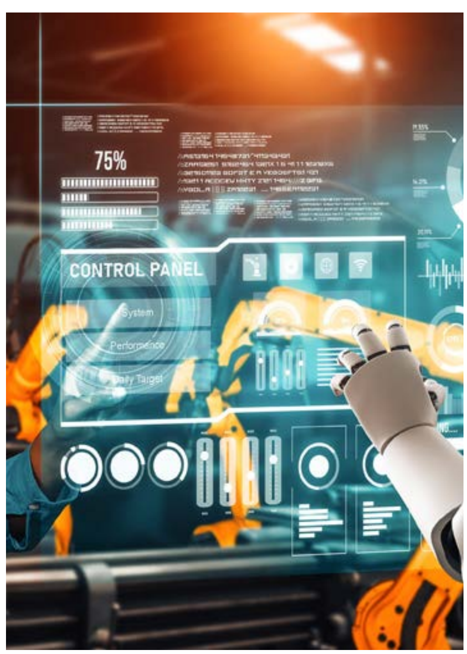
Although AI is rapidly advancing, true autonomy only appears when a system can sense, decide, act, and learn in a governed loop. In operations, governance matters as much as optimization. The addition of AI doesn't change existing validation, auditing, and compliance requirements. An upfront assessment turns unknowns into measurable signals and defines the policies, controls, and thresholds AI can act within (with HITL escalation when needed). From there, AI can both prioritize where automation will pay off and increasingly drive the automation itself, with audit trails and decision logs that make outcomes explainable, compliant, and repeatable.

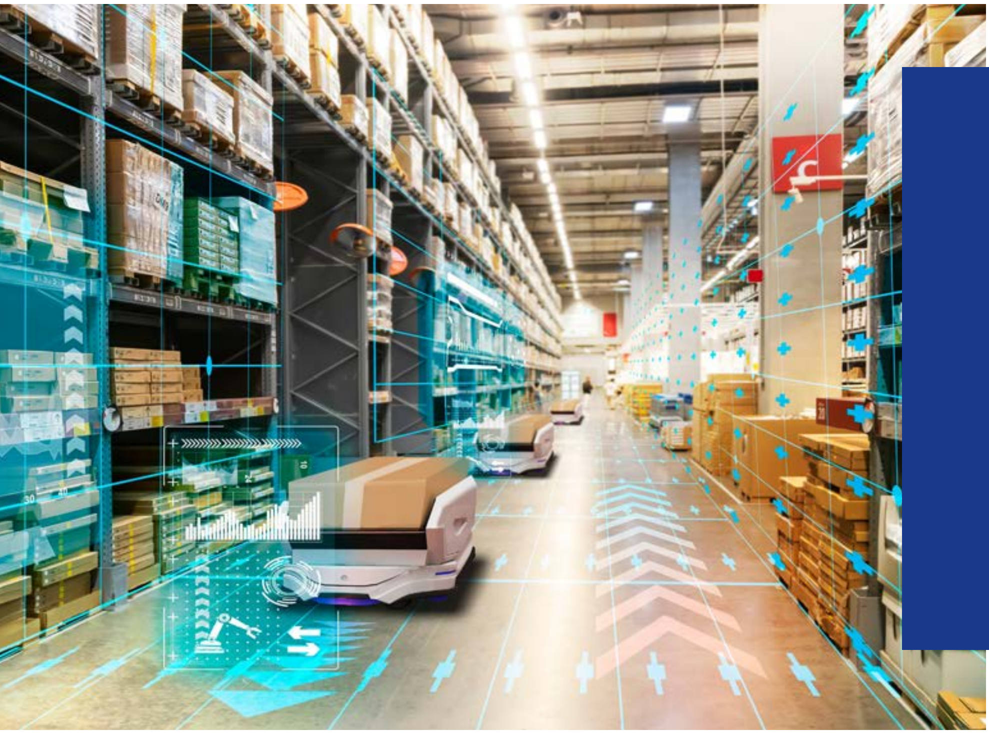
AI IS THE FUTURE OF SUPPLY CHAIN—AND THE FUTURE IS NOW

Agentic AI has drawn enterprise attention with the promise of autonomous operation and intelligent execution. Gartner predicts that by 2028 15% of day-to-day work decisions will be made autonomously through agentic AI (up from 0% in 2024), and that 33% of enterprise software applications will include agentic AI within the same timeframe (versus less than 1% today). On the other hand, Gartner also notes that more than 40% of agentic AI projects will be canceled by the end of 2027[1]. Meanwhile, robotics advancements are making robots more accessible and capable[2].

While these recent advancements can unlock untold value and benefits, organizations that focus on optimization and management of AI capabilities and automation will start to separate themselves from the pack. An example of this exact orchestration follows on the next page. The solution features an integrated architecture that automates complex, multi-step warehouse operations with unprecedented efficiency by unifying fragmented automation into a scalable, intelligent ecosystem. This case demonstrates how modern AI-driven orchestration can unlock the full value of warehouse robotics and deliver outsized financial and operational returns.







SHARED ROBOT BRAIN: CUTTING COST PER BOX MOVED BY 85%

Situation: A live retail distribution operation for the second-largest U.S. retailer needed a better way to execute high-volume, variable cross dock warehouse workflow without scaling labor linearly. It needed a shared robot brain that could coordinate people, robots, and decisions across a dynamic, high-variability warehouse workflow. Traditional automation and point solutions were not built to coordinate changing work across people, robots, and real-time operational constraints. The operation needed a system that could reduce cost per move, improve responsiveness on the floor, and create a repeatable blueprint for scaling automation across sites.

Action: Retailer's 3PL worked with Destro AI to deploy a shared intelligence architecture built on MothershipOS and VisionOS. Instead of managing each robot or task as a separate point solution, MothershipOS served as the central decision layer, continuously determining what should happen next, assigning work dynamically, and coordinating execution across the operation. VisionOS added perception capabilities to autonomously count cartons and verify SKU quantities, reducing manual scanning and verification steps. The implementation was designed to support multiple workflow types through one intelligence layer rather than requiring separate software stacks for each use case.

Result: The deployment created a path to reduce cost per box move from \$0.20 to \$0.03 per case. It showed an estimated payback period of 2 months, a projected headcount reduction of 72 per warehouse, and approximately \$9 million in projected savings per warehouse over 3 years. Beyond the headline economics, the retailer gained a more flexible operating model: one that can adapt to changing workflows, coordinate many robots through one brain, and extend automation into environments that have historically been too dynamic and operationally messy for conventional systems.

This case illustrates how agentic AI can move beyond isolated pilots and become an operational control layer for real warehouse execution. By combining shared brain, agentic AI, real-time workflow coordination, and embodied AI, this solution enables enterprises to automate complex workflows with fewer point systems, faster payback, and a clearer path from pilot to scaled deployment.

AI'S IMPACT ON THE SUPPLY CHAIN WORKFORCE

This year's survey results also show strong belief in the impact of AI, with nearly two-thirds of respondents (62%) rating its future impact on supply chains and material handling as either significant (34%) or transformational (27%). Conversely, only 6% of respondents believe AI will have a limited or minimal impact on the future of supply chain and material handling (Figure 4).

As reluctance shrinks and organizations increasingly weave AI into their core operations, the focus shifts from adoption to scaling: operationalizing what works, connecting AI across silos, and upskilling teams. Two survey themes underscore the operating pressure and disruptive urgency today's leaders face. First, AI and automation are becoming central to the supply chain agenda, with AI expected to be the most disruptive technology of the next decade (followed closely by robotics and automation). Second, talent is the top operating pressure, with workforce shortages becoming a major operating constraint.

The top concern for more than a third of employers is equipping workers with the skills and knowledge they need to maximize the potential of smart manufacturing and operations. One way to achieve this is by investing in upskilling partnerships. For example, GXO partnered with a UK university to have its employees participate in an apprenticeship program that offered hands-on robotics training, which is integral to GXO's operations[3]. Another large supply chain company partnered with a university to help frontline team members develop new skills and broaden their expertise[4]. Organizations should consider implementing upskilling practices that directly engage employees in critical operating activities, such as constructive feedback provided to employees performing overrides after shifts, simulation drills before changeovers, or data interpretation before shift start-up.

Continuous learning will help employees adapt to new technologies, bridge skills gaps, and boost morale, while organizations will see their service quality and ROI increase. Also, Harvard Business Review reports that this type of upskilling investment can enhance retention and create an overall work experience that is more fulfilling[5].

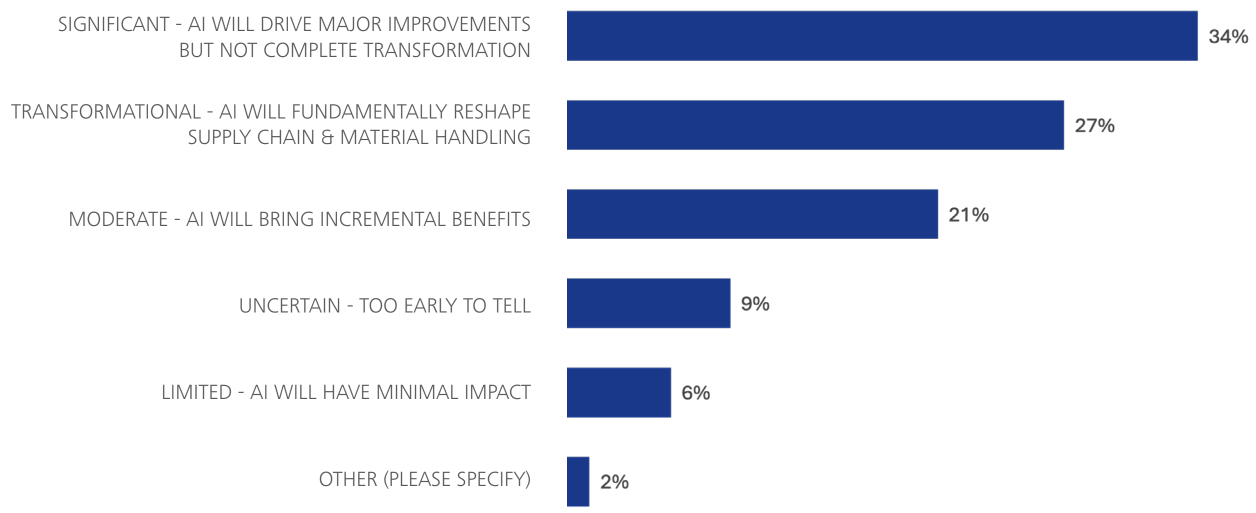


Figure 4: Survey results – expected impact of AI

AI-DRIVEN SHIPMENT CREATION

Situation: In freight operations, shipment creation (also called routing or load building) is a high-stakes process. Every misrouted shipment, delayed appointment, or overlooked hazmat detail creates downstream costs and erodes customer trust.

Capacity LLC's operators faced especially complex workflows that consumed time and created service risks. Human operators were tasked with consolidating multiple purchase orders into single shipments; applying early/late ship-date rules with narrow delivery windows; reconciling mismatched data across spreadsheets and portal records; scheduling appointments in multiple third-party freight portals; and reviewing and interpreting hazmat data sheets for compliance. These tasks often required manual rework, slowing execution and increasing the risk of errors.

Action: Ventus AI deployed a shipment creation and routing automation agent tailored to Capacity's workflows. The agent parses spreadsheets and validates order details; consolidates shipments automatically according to routing rules; applies custom early/late ship windows; creates shipments directly in E2Open, Uber Freight, and MercuryGate; schedules appointments across third-party portals; parses hazmat sheets, flagging critical requirements instantly; and returns clear exception messages (e.g., missing POs, mismatched data).

Result: Ventus AI's automation delivered measurable impact within weeks. Shipment creation time has been reduced from ~5 minutes per load to creating shipments in real time. Routing errors are reduced, lowering costly service failures. Hazmat documentation is processed instantly for improved compliance. Appointments are scheduled automatically with zero operator input. Perhaps most important, operators now have more time to focus on exceptions and higher-value customer service.

Through Ventus AI, Capacity has reduced costs, eliminated repetitive tasks, and strengthened customer trust. "Ventus AI gives our team superpowers. It's helping us automate complex tasks like setting appointments across third-party portals and parsing and understanding hazmat data sheets—freeing up our people to focus on higher-value work and provide better services for our customers," says Jeff Kaiden, CEO of Capacity LLC

Capacity's success highlights how leading logistics providers are rethinking back-office operations with AI:

- Smarter automation—not just faster, but more accurate and compliant
- Scalable AI agents—enterprise-ready tools that work across multiple freight systems
- Customer-first logistics—allowing operators to focus on service, not data entry

This is more than an isolated case study. It's proof that AI in freight operations can handle complex workflows at scale, from shipment creation to hazmat compliance.

THE EXECUTION GAP—WHY MOST AI PILOTS DON'T SCALE

Despite survey results showing broad interest in AI, enterprises are encountering significant obstacles translating pilots into production-ready solutions. Deloitte's 2026 Emerging Technology Trends study notes that while 30% of surveyed organizations are exploring agentic options and 38% are piloting solutions, only 14% have solutions that are ready to be deployed and just 11% are actively using these systems in production[6]. Also, 42% of organizations report they are still developing their agentic strategy roadmap, with 35% having no formal strategy at all.

Most leaders aren't blind to AI's potential; they are getting stuck on where to start and what it takes to scale. The barriers are real and practical—unclear use cases and automation cost, paired with limited understanding; difficulty building business cases; talent shortages; and budget constraints.

DESIGNING PILOTS FOR SCALE

You can't out-execute a bad design. Thus an operational assessment of your system(s) and organization is key. Organizations need a disciplined alignment on outcomes and constraints (service, cost, risk), paired with a structured review of current workflows, system touchpoints, decision rights, and baseline performance against existing key performance indicators (KPIs). Before starting a pilot, it's essential to clearly define what decisions are affected, who owns the process, and how success will be measured in a regular operating environment. Without that, a pilot might look great on paper but its broad applicability and scalability remain a question.

Closing the execution gap requires shifting from standalone proofs of concept to a repeatable decision-and-execution loop so the model is not only accurate but also becomes the standard way to get work done. Each use case should be designed around a four-phase approach of:

1. Sense
2. Decide
3. Execute
4. Learn

A practical closed-loop model connects assessment findings to automation outcomes.

Identify critical KPIs and verify the quality of the data. Translate those verified signals into a bounded set of decisions with HITL guardrails and full auditability. Execute through existing workflows, using automation where effective, and capture full telemetry to validate outcomes and drive continuous improvement through learnings.

The decision-and-execution loop becomes the unifying architecture that scales across vendors, sites, and functions without reinventing the stack each time. A modern fulfillment operation already has data, tools, dashboards, and increasingly sophisticated models. What's often missing is coherent coordination. An orchestration layer that converts diverse signals into bounded, explainable decisions drives consistent execution across systems and sites—while learning fast enough to keep pace with reality.

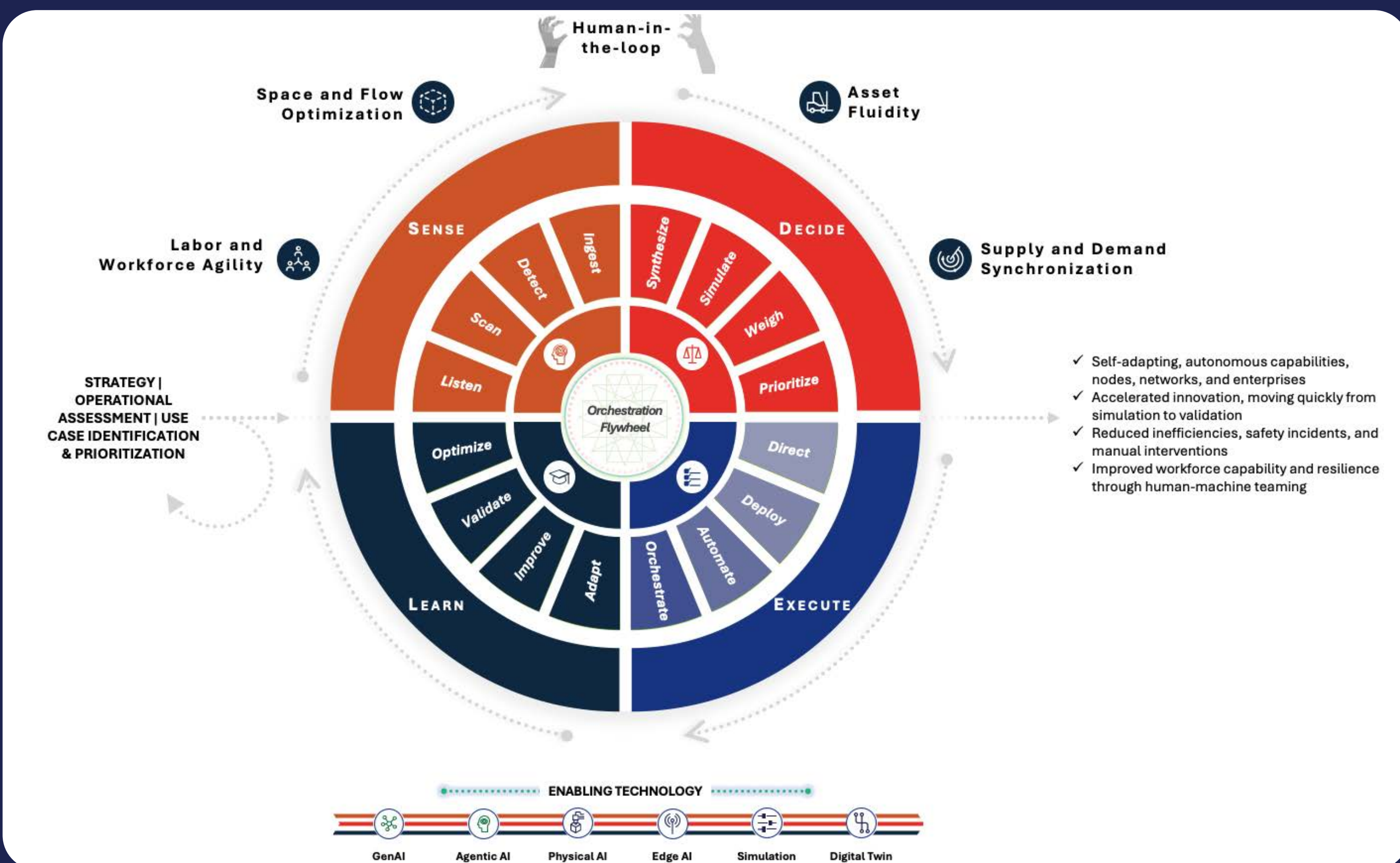
LOOKING AHEAD

The integration of generative AI, agentic AI, physical AI, and edge AI (the deployment of AI algorithms and models directly on local edge devices such as IoT sensors, smartphones, autonomous vehicles, and embedded systems rather than a centralized cloud) into operations is ushering in a future where fulfillment and supply chain activities become software-defined, perpetually adaptive, and backed by intelligent orchestration engines. This transformation does not just improve what already exists, it fundamentally changes how challenges are solved, how capital is invested, and how workforce capacity is harnessed.

ORCHESTRATION IN ACTION – THE AI AUTOMATION PLAYBOOK

When harmonized, supply chain technologies and AI can function as an intelligent control layer across facility, supply chain, and labor networks—closing the loop between virtual prediction and real-world execution. Agentic AI can draw on digital twin data to run simulations and recommend routing, fulfillment, and resource allocations. Physical and edge AI can capture real-world signals and execute decisions, while human teams engage through clear interfaces and validation points to review, validate, and iterate.

This orchestration and automation playbook is built on four key pillars: (1) dynamic supply and demand synchronization, (2) asset fluidity and predictive maintenance, (3) space and flow optimization across the network, and (4) labor and workforce agility—all governed and optimized with human-in-the-loop collaboration.



Respondents to this year's survey state that enhancing demand forecasting and inventory optimization is the top application for AI (cited by 33%). This process continuously balances demand, inventory, and order priorities so volatility becomes a managed signal. The flywheel helps sense shifts early, decide tradeoffs quickly, execute reallocations, and learn which signals best protect service and cost.

Asset fluidity and predictive maintenance enables adaptive automation and treats equipment, automation, and transport capacity as re-allocatable nodes that can be rerouted or "hot-swapped" when constraints are sensed. According to this year's survey, 30% of organizations apply AI to predictive maintenance and 26% apply it to optimizing logistics and transportation routes.

Space and flow optimization enables organizations to move from static slotting and layouts to responsive slotting, zoning, and buffer logic that optimizes throughput and enhances safety. Digital twin/simulation lets teams test changes virtually, execute with confidence, and learn which flow patterns reduce congestion, bottlenecks, wasted travel, and location visits.

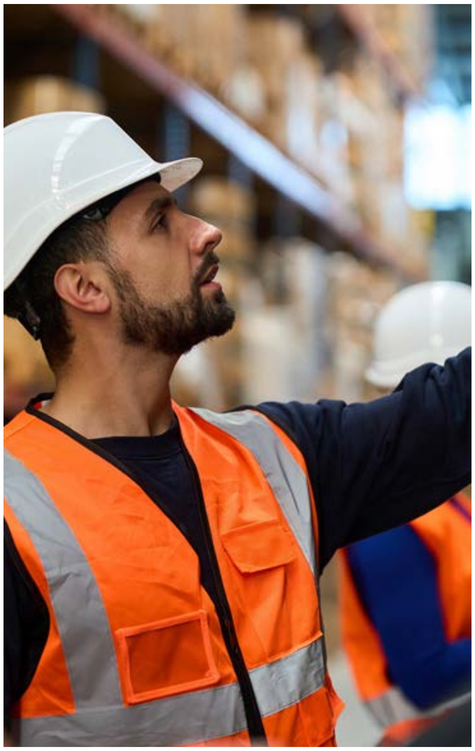
Human-in-the-loop (HITL) collaboration is used throughout the cycle so autonomy scales safely. Automation handles routine tasks, humans approve/override high-impact exceptions, and everything is auditable. HITL converts exceptions into learning, strengthening trust and accelerating continuous improvement. This symbiosis translates to improved labor and workforce agility, including higher engagement, less burnout, and a culture of continuous improvement.

Each of these pillars can independently drive value; however, the full ROI comes into play when all are working in concert, bringing network orchestration and automation to life through the four phases outlined previously: sense, decide, execute, and learn.

Imagine an inbound reefer load at risk of a temperature excursion due to travel delays. In this example, AI agents coordinate to protect shelf life and customer value. In the sense phase, the system combines cold-chain telemetry, live ETAs/traffic/weather, DC dwell signals, and store-level demand to flag quality risk. In the decide phase, it then synthesizes and simulates options (hold course, expedite, or re-route) and recommends re-routing to a closer distribution hub—but only because asset fluidity exists (i.e., the truck can be dynamically re-dispatched with compliant routing and updated appointments). An HITL checkpoint approves the reroute and automatically triggers aligned actions and notifications to the DC director/manager and freight forwarder/carrier so the plan is executable, auditable, and controlled.

In the execute phase, the closer distribution hub confirms its capacity and uses space & flow optimization to make room immediately: inventory is dynamically re-slotted, cross-dock lanes are reserved, and the warehouse control layer pushes clear instructions/objectives to robots for receiving and automated unloading (with objectives to prioritize quality-critical handling, preserve FEFO/lot traceability, and stage straight to outbound so product is customer-ready upon arrival). This entire scenario is further enabled by labor and workforce agility: leadership receives consistent signals to recommend optimal staffing across each node, with additional HITL approvals for any high-impact overrides.

In the learn phase, the system validates outcomes–dwell time, temperature stability, quality, service–and updates thresholds, playbooks, and automation rules so the next reroute happens faster with fewer manual interventions. The feedback loop also informs the operational assessment and use case prioritization, providing key metrics and insights to inform future improvements (e.g., what caused the temperature excursion and how to prevent it).



Automation technologies—robotics, AI, AGVs, and wearables—are transforming warehouse and manufacturing operations. To deliver real value, organizations must align these investments to clear goals and a deliberate deployment strategy.

Fred Cox, Director of Manufacturing at Central Shops, Disney

THE AUTOMATION DIVIDE: FROM MANUAL OPERATIONS TO SCALABLE AUTOMATION

Physical automation technologies are reshaping warehouse and manufacturing operations. Autonomous vehicles, robots and cobots, advanced conveyance systems, AI, and the Internet of Things (IoT) are changing how companies fulfill orders and make products, enabling faster delivery, more personalization, and greater flexibility as demand shifts. Meanwhile, the pace of change is accelerating, making it critical for organizations to continuously evaluate their operations and invest in the tools that will keep them competitive.

Implementing automation isn't simple. The journey from design to go-live is long and complex, and selecting the right solution can be difficult. To thrive, organizations need both the right technologies and a disciplined approach to decision-making and continuous improvement.

Automation is no longer optional, and it's not a one-time effort. Let's examine how automation is rewiring operations and enabling new business models—and why a practical playbook for implementing automation is essential to achieving the expected results.

KEY DRIVERS OF WAREHOUSE AUTOMATION

According to the 2025 State of Smart Distribution Study, 94% of companies say supply chain disruptions are critically or moderately impacting their organization. To mitigate these challenges, respondents to our survey are making strategic investments to transform their warehouse and manufacturing operations—with a priority on reducing operational costs (71%), improving the customer experience (51%), improving delivery times (48%), and addressing labor/talent shortages (44%) (Figure 5).

Labor-intensive manual workflows and purely human workforces are no longer sufficient to meet the needs of today's customers. As figure 6 shows, companies are moving aggressively towards a future based on smart technologies such as AI (47%) and advanced analytics (38%) to enhance their operations and achieve the flexibility and scalability required to address customer needs and changing market conditions. Growth rates are expected to be particularly high for

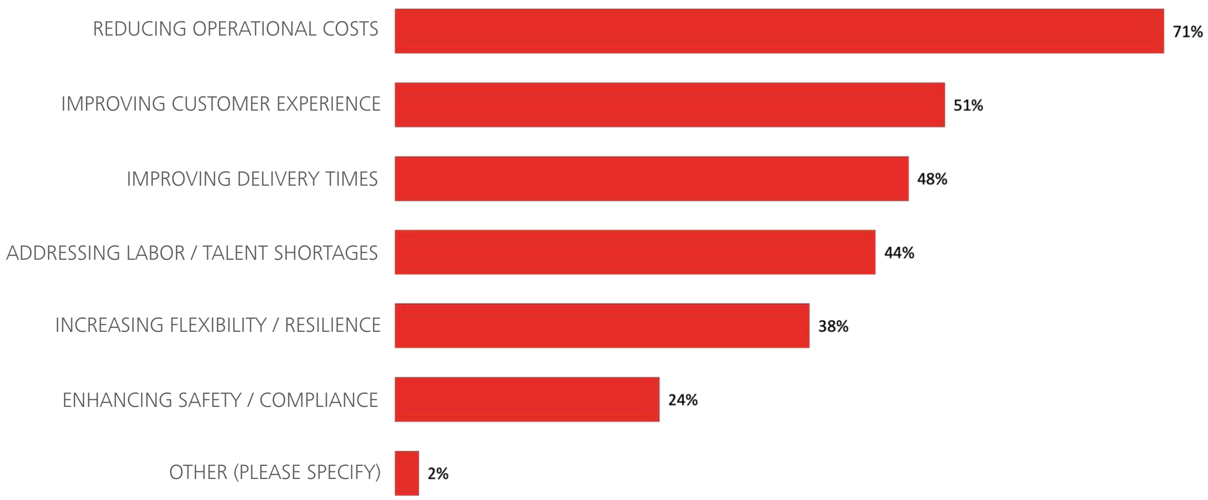


Figure 5: Survey results – strategic investment priorities

physical AI technologies, including humanoid robotics (from 4% to 28% over the next five years), autonomous vehicles/drones (from 16% to 34%), and wearable/mobile technology (from 28% to 41%), which will likely be the front-end user interface for many of tomorrow's physical AI systems.

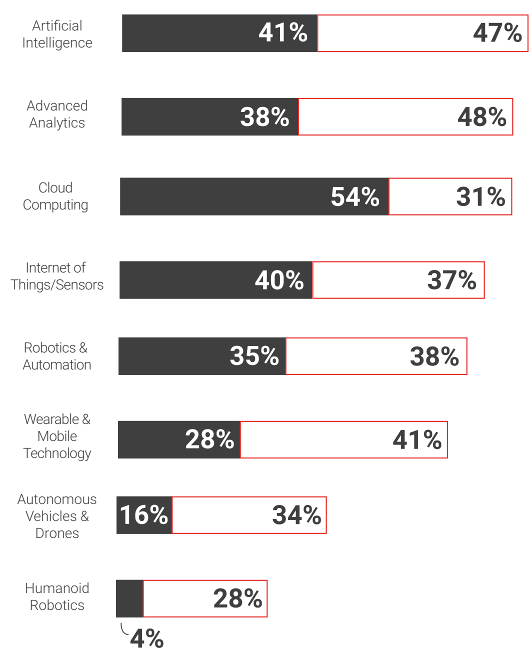


Figure 6: Survey results – adoption of advanced technologies

These technologies—when combined and working together in a cohesive next-gen facility—can unlock unprecedented levels of efficiency, accuracy, and responsiveness. Warehouses are transformed into intelligent hubs where real-time data flows seamlessly, enabling predictive maintenance, dynamic inventory management, and rapid order fulfillment. On the manufacturing side, connected processes and adaptive robotics allow for greater customization and shorter production cycles, all while enhancing workplace safety and sustainability.

COMPARING AUTOMATION SOLUTIONS

Over the next five years, automation technologies are expected to advance by leaps and bounds. To understand which solutions best fit your needs, it's important to understand the pros and cons—and how each technology is evolving. Here's a look at some of today's most commonly implemented solutions:

- AI-powered smart warehouse management systems (WMS) and advanced analytics can improve planning and decision-making across slotting, labor planning, wave/batch optimization, and replenishment, but they require strict master data governance to ensure decisions are based on accurate information.

- Modular warehouse control systems (WCS) can connect and help manage automation solutions alongside WMS, ERP, and MES solutions.
- Robotics and automation can reduce labor risk while optimizing space and increasing throughput, but they typically involve high capital expenditure (CapEx), significant ongoing maintenance, and added engineering complexity.
- Automated guided vehicles (AGVs) can streamline low-value travel, scale with demand, and improve safety, but they require disciplined staging and labeling, introduce process constraints, and require effective fleet-management orchestration.
- Wearable technologies can boost productivity, simplify training and task guidance, and improve order accuracy, but may raise ergonomics/comfort concerns and often rely heavily on network/radio frequency (RF) performance.
- Humanoid robots have the potential to fit more easily into existing layouts and take on diverse tasks, but the technology is still immature and less proven than AGVs and autonomous mobile robots (AMRs).

However, some humanoid robotics companies are showcasing their ability to deliver value today, with one example being VersaBot. A landmark innovation of VersaBot is its elimination of LiDAR sensors. Instead, it relies solely on the company's self-developed 3D vision sensors and AI algorithms to complete navigation, obstacle avoidance, grasping, and other core functions. This pure-vision system captures rich visual data, enabling the robot to better understand complex and changeable environments, adapt to diverse task demands, and deliver enhanced capabilities like being able to handle out-of-stock scenarios – the robot's dynamic perception system detects the shortage and promptly feeds back an order exception alert.

As organizations strive for coordinated, end-to-end automation, they must avoid traditional integration silos. An example of a solution that is delivering a coordinated, end-to-end automation experience is HmWare by HUBMASTER. HMWare® functions as a native AI-enabled WMS and WCS layer for the HUBMASTER ASRS, managing storage strategies, task sequencing, and throughput optimization through configurable software logic. AI-assisted execution enables the system to dynamically adjust priorities, routing, and resource allocation based on real-time system state and operational demand. A key differentiator of this native architecture is its ability to manage ASRS automation and mobile robotics, including AGVs and AMRs, within the same control platform. By embedding vision, AI, and multi-technology orchestration natively into the HUBMASTER ASRS, this IT innovation delivers a software-defined automation platform that is more reliable, adaptable, and scalable than conventional add-on control architectures, enabling advanced warehouse solutions that were not previously feasible.

UNDERSTAND WHERE YOU ARE, WHERE YOU'RE GOING, AND WHY

Automation requires significant investment, so it's important to clearly understand why you're pursuing it. According to the Supply Chain Management Review Agility At Work survey, most companies approach robotics and automation guardedly: 38% are cautious yet practical, 24% are slow movers, and 13% prefer to wait and see, compared with 15% that are early adopters and 10% that are innovators[7]. Today, economic conditions might be making companies hesitate; however, the data suggests many organizations see enough potential value to get started sooner rather than later.

An operational assessment can clarify your organization's current maturity level and specific goals—whether you're trying to increase throughput, reduce labor dependency, improve order accuracy, or boost agility. By defining clear priorities (e.g., cost, safety, sustainability, service), leaders can align investment, change management activities, and technology selection around a shared vision. The assessment should provide an objective look at strengths, gaps, workflows, and current key performance indicators (KPIs) to pinpoint what problems automation aims to solve. It's also important to determine what matters most. Some organizations prioritize flexibility and scalability; others prioritize quality or accuracy over speed. There is no one-size-fits-all approach. Leading organizations mix and match technologies to build next-generation operational capabilities that fit their unique needs and create enduring competitive advantage.

FULL-SCALE TRANSFORMATION OR PHASED APPROACH?

Once you understand which technology solution(s) your organization should invest in—and why—the next step is choosing a deployment strategy: either full-scale transformation, or a phased approach.

Many companies underinvest in the deployment strategy behind their technology decisions, treating automation as a near-term purchase rather than a multi-year capability build. A recent Hai Robotics survey suggests planning horizons are typically short: 53% of companies plan only 1–2 years ahead, 33% plan 3–5 years, and just 3% plan beyond 5 years[8].

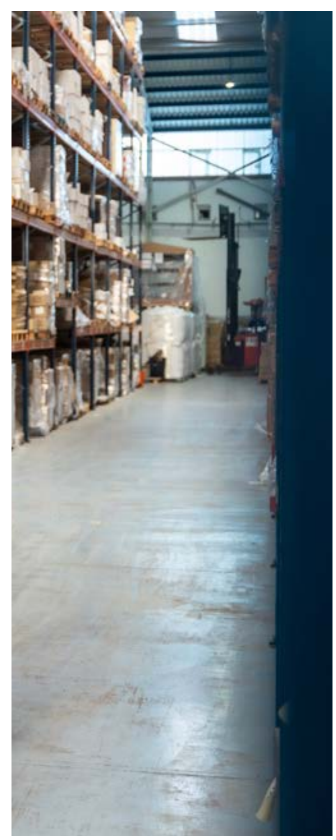
In theory, full-scale transformation can unlock end-to-end efficiency gains more quickly; however, it requires significant upfront investment, robust change management, and disciplined risk mitigation. As such, many companies prefer a pilot-first approach that builds toward fully scaled automation through incremental deployment, delivering focused quick wins while reducing operational risk and disruption.
Key decisions include site rollout plans, integration architecture, operating model changes, and governance—all of which need to be carefully planned and thought out as they may require a longer runway than the hardware or software itself.

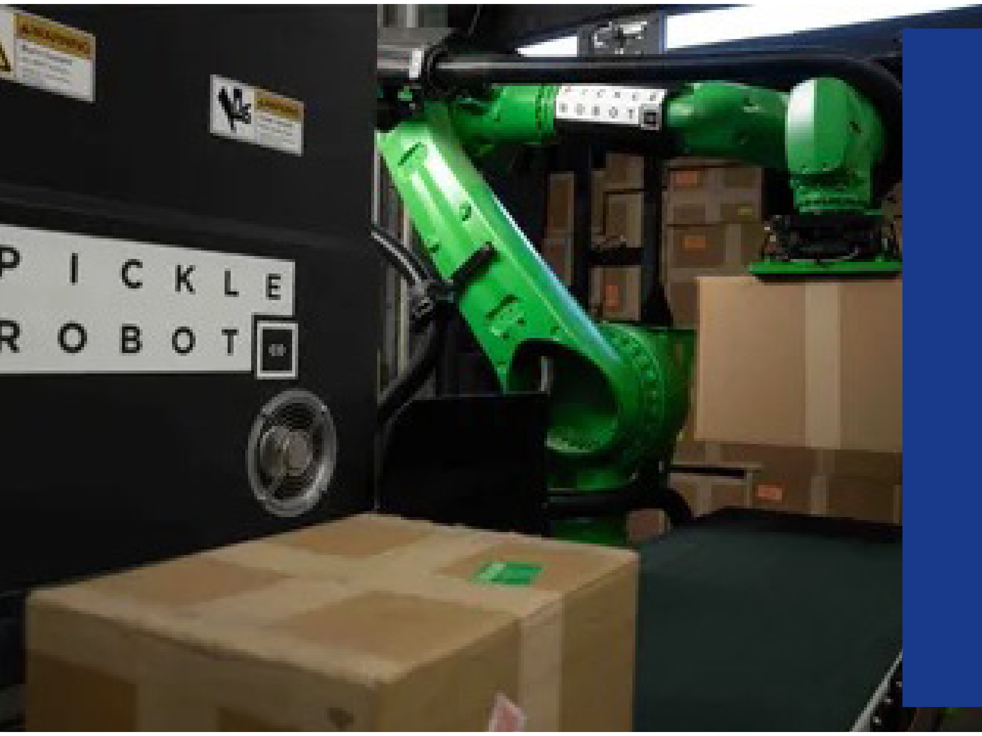
Regardless of the deployment strategy, success depends on integrating new solutions with legacy systems (e.g., warehouse management systems (WMS), programmable logic controllers (PLCs), and existing databases). Rather than replacing everything, organizations should conduct system audits, define integration points, prioritize compatibility, and partner with vendors who can bridge legacy and modern environments to minimize downtime and protect prior investments.

LOOKING AHEAD

Successful warehouse and manufacturing automation doesn't end at go-live—it marks the start of a continuously evolving innovation lifecycle. Leaders differentiate themselves by embracing an approach of ongoing, data-driven improvement, using insights from automated systems and predictive analytics to spot bottlenecks, refine processes, and respond to shifting market needs before performance is impacted.

As automation technology advances, physical and digital operations will become even more integrated, enabling more responsive supply chains and scalable, personalized manufacturing. Capturing lasting value will require investing not only in technology, but in people as well: developing digitally enabled problem-solvers and fostering a culture of experimentation, learning, and cross-functional collaboration. Ultimately, organizations that embrace AI, analytics, and a holistic innovation approach won't just adapt to the future—they'll help shape it.





ROBOTIC TRUCK UNLOADING

Situation: A leading global manufacturer of power tools, outdoor power equipment, and floor care products operates regional distribution centers in major logistics corridors where competition for qualified staff is highly competitive. At many of their facilities a significant volume of inbound inventory arrives via sea-going import containers that require manual unloading. The work is hot in the summer, cold in the winter, and always physically demanding. As such, the organization's inbound docks had the highest turnover rates and worker compensation claims of any department in the warehouse. To unload containers, the company has used various combinations of company employees, temporary labor, and lumping services, but regardless of the approach, staff reliability, cost, and safety issues associated with the inbound process have been a constant challenge for site management, who need to ensure employees have a safe and fulfilling work experience—and that operations run smoothly to meet customer commitments.

Action: The company deployed trailer-unloading robots from Pickle Robot to automate container unloading at multiple distribution centers. Product packaging to be handled includes both brown 'master carton' type boxes and display boxes (sometimes referred to as 'ship in own carton' or SIOC freight), and box counts range from 300 to 3,000 packages per container. The Pickle robots unload boxes as small as 5" cubes and as large as 24" x 24" x 30", weighing up to 50 pounds. Each robot serves 2 - 4 adjacent dock doors and processes freight at 400 to 900 pieces per hour depending on the size and weight of the products. The distribution centers typically operate the robots 2 shifts per day and 5 to 6 days per week depending on inbound volume, which fluctuates seasonally.

Result: The distribution centers have reduced team turnover within the inbound departments by 42% and some full-time employees have asked to be re-assigned back to inbound operations to work with the robots. Today, the Pickle robots process 30 - 65% of inbound import freight depending on the seasonal mix. The robots do not unload the heaviest products, such as ride-on lawnmowers, but they do handle items as large as push mowers and table saws. The average container unload time is about two hours, and the distribution centers report both a better safety record and lower month-over-month costs compared to the previous fully manual process.

Those who connect operational excellence, AI-driven orchestration, and workforce readiness into a single playbook will not just withstand disruption; they will convert it into sustained performance and growth.

Wanda Johnson, Supply Chain Technology Fellow, Deloitte Consulting

FUNDING WHAT SCALES: TURNING INNOVATION INTO OPERATIONAL RESULTS

After a sharp spending surge in 2022 and 2023, supply chain investment levels have returned to a steadier baseline. The year-over-year trend shows weighted average planned spend peaking at $21M (2022) and $26M (2023), then settling to $13M (2024) and $11M (2025)—back in line with pre-COVID patterns (generally $7M–$14M from 2015–2021) (Figure 7).

This normalization is not a pullback from innovation; instead, it is a shift to value realization. Many

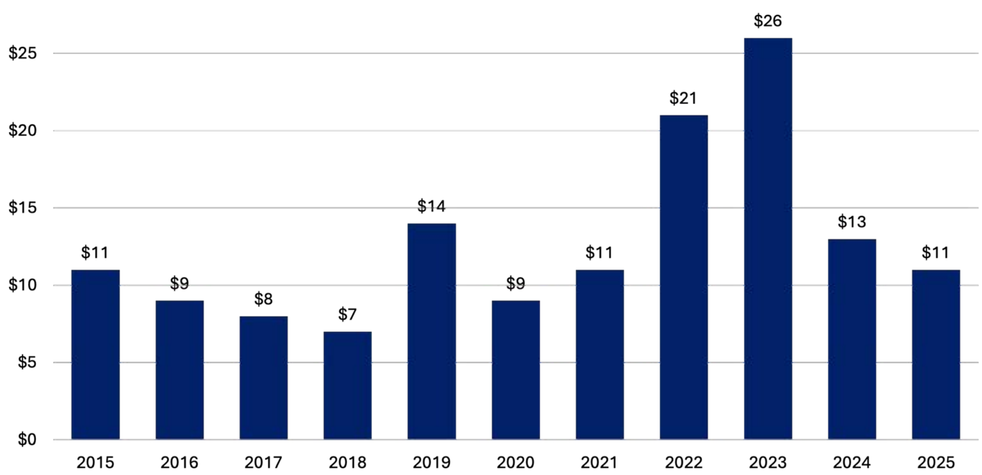


Figure 7: Survey results – supply chain innovation investment

organizations made foundational bets during the surge and are now moving into the operational phase of transformation—integrating platforms, hardening data and security, scaling adoption, and improving day-to-day execution. The opportunity for 2026 and beyond is to pair a disciplined operational focus with a longer-horizon plan for the next wave of technology, including AI, robotics, and drones, so that investments compound instead of fragment.

Near-term spending expectations also skew toward targeted, ROI-driven programs rather than mega-transformations. This distribution of estimated investment supports what many organizations are experiencing right now: capital is available but is increasingly gated by measurable payback, clear risk controls, and scalable operating models (Figure 8).

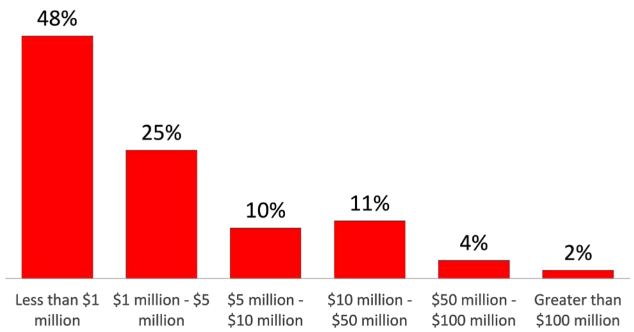


Figure 8: Survey results – supply chain investment over the next two years

SHIFTING FOCUS FROM NOVEL TECH TO RIGHT TECH WITH MEASURABLE VALUE

The normalization in spending reflects a more disciplined investment approach: companies are stepping back to confirm what problem they're trying to solve, selecting the right use case for the bottleneck, and looking for a credible path to benefits before scaling. Instead of funding broad, tech-led programs, leaders are aligning automation and AI with operational goals and building the controls required to scale effectively and safely. Investment decisions are increasingly anchored by identifying the operational constraints that need to be resolved and then selecting a relevant use case with measurable outcomes, rather than deploying "cool tech" in search of a problem (Figure 9).

According to this year's survey, 56% of organizations expect to increase their spending on supply chain innovation. Leaders are treating innovation funding like a risk-managed portfolio where the default strategy is phased implementation with pilots and proof-of-value gates. This approach can reduce ROI uncertainty and operational disruption. It also forces early clarity on integration, data readiness, cybersecurity, safety, and operating models to ensure the solution is deployable beyond a single site.

Program teams are spending more time upfront on baseline metrics, process maps, SKU/order profiles, exception patterns, and other crucial factors that determine whether a solution will actually perform in production.

This increased investment discipline also reflects an evolving leadership mandate. Gartner's 2025 Supply Chain Symposium messaging emphasizes that Chief Supply Chain Officers (CSCOs) are being measured less as logistics operators and more as enterprise integrators who are orchestrating complex, cross-functional solutions

Figure 9: Survey results – primary driver for technology investments

that drive customer satisfaction, brand trust, and revenue growth[9]. Also, as AI takes on more day-to-day decisions, the CSCO's focus is shifting toward governance, risk management, and scenario planning to better justify, control, and scale technology investments[10].

INVESTING FOR THE FUTURE

Investing over the next two years is less about returning to "big bang" transformations and more about directing significant capital to the capabilities the market is prioritizing. According to our survey, future investment is shifting towards artificial intelligence and robotics & automation as organizations increase their focus on automating both decisions and execution at the point of work (Figure 10). However, these solutions will only scale if leaders also fund enablers such as cloud computing, IoT, and the data controls necessary to make AI trustworthy, such as security, integration, and governance.

A winning approach will likely look like this: AI and analytics tied to a few key use cases with solid baselines; automation where labor, safety, and throughput benefits are measurable; and a robust data, platform, and cybersecurity foundation—all supported by workforce upskilling to keep adoption on pace with deployment. By investing this way, companies can convert innovation spending into tangible performance improvements such as faster decisions, safer operations, and more resilient service.

SECURING LEADERSHIP APPROVAL AND FUNDING

Obtaining leadership approval for warehouse automation requires translating operational benefits into a compelling case for investment. This is especially important for automation, where ROI uncertainty can be a major barrier. The most effective financial proposals highlight payback, risk controls, and measurable outcomes.

Key features of an executive-ready business case:

- The "before" snapshot—a data-driven baseline that reflects current-state process maps, throughput, error rates, labor cost, safety incidents/near misses, SKU/order profile, etc.
- The "after" vision—quantified value that outlines hard-dollar benefits (e.g., labor productivity, reduced damage/returns, fewer expedited shipments, improved space utilization) and risk-adjusted benefits (e.g., safety, compliance, resilience)
- Clear ROI that considers total cost of ownership (TCO), payback period, and sensitivity ranges
- Comparison of alternatives such as AMRs vs manual labor, forklifts, conveyors, or automated storage and retrieval systems (showing why the chosen solution fits the company's order profile and constraints)

Stakeholder-specific framing:

- CFO/Finance. Focuses on factors such as ROI, TCO, payback period, depreciation assumptions, and downside protections.
- Operations/Plant Leader. Emphasizes factors such as throughput, cycle time, downtime risk, and labor stability.
- Safety/Health Leader. Highlights factors such as risk reduction, ergonomics, incident reduction, and insurance implications.
- IT/Engineering. Focuses on factors such as integration with WMS and ERP, cybersecurity, and operational support models.

Every investment request should include an executive summary, phased timeline, and the ramifications of doing nothing. Identify benefits such as labor cost savings, increased throughput, and improved safety. Then use a pilot to prove the business case, converting the investment decision from a leap of faith into a controlled capital allocation.

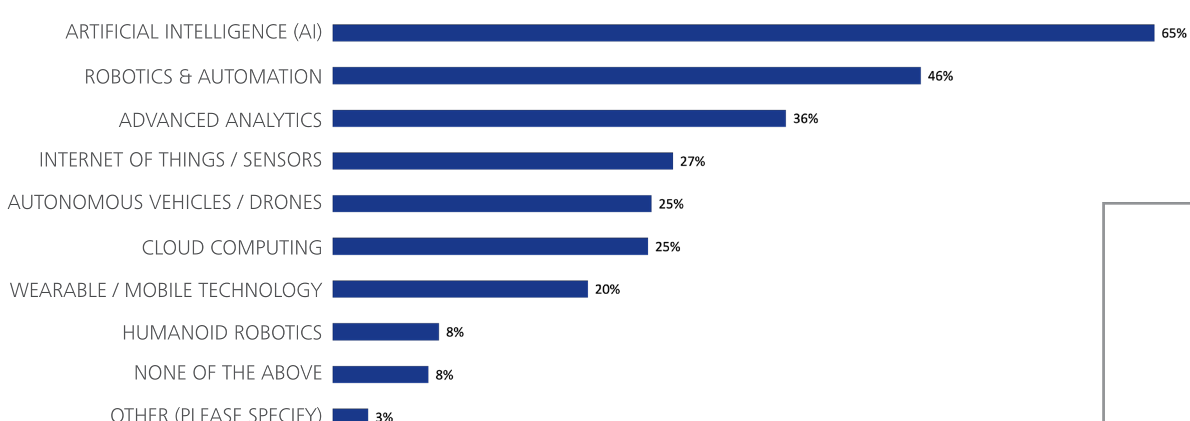


Figure 10: Survey results – planned technology investments

Supply chains are chasing many 'shiny objects,' but technology alone isn't enough. Integrating automation with workforce strategy is essential for lasting advantage.

Stephanie Thomas, Associate Professor of Practice of Supply Chain Management Sam M. Walton College of Business, University of Arkansas

LAYING THE FOUNDATION FOR AI & AUTOMATION

Rewiring a supply chain for the future requires viewing the operation as an integrated system (process, inventory, technology, and workforce) and building an evidence-based view of what is currently constraining throughput, accuracy, and agility. When organizations do this well, they stop debating symptoms and start prioritizing the improvements that unlock the most value.

Here are some practical actions organizations can take now to harness the power of AI and automation and set the stage for an AI-orchestrated supply chain.

ASSESS THE CURRENT STATE

Without an accurate understanding of where you are today, you can't make viable plans about where you want to go—and how to get there. A comprehensive warehouse or network assessment establishes a fact base across four key dimensions: process flow, labor utilization, technology integration, and inventory practices.

An end-to-end examination of your current operational reality lets you pinpoint bottlenecks, identify existing problems, and learn how your best talent is using innovation to work around the constraints of your existing systems and processes.

Operational deficiencies often boil down to a quantifiable set of constraints (e.g., lack of resources, inadequate systems, poor alignment). With a verified list of such constraints, you can begin optimizing systems and processes to enhance existing points of excellence and smooth out identified points of friction.

Assessments need to separate system capabilities from execution reality. For example, two sites can run the same WMS and still produce very different inventory accuracy because the drivers are often operational: whether directed work is executed consistently, tasks are closed out correctly, and exceptions are handled the same way every time when the system and the floor do not match.

USE INNOVATIVE TOOLS TO PRIORITIZE OBSERVATIONS INTO DECISIONS

The latest tools can accelerate insight by making constraints visible and testable. Used properly, they can help leaders answer critical questions, such as: Where is work actually getting stuck? Which changes will improve flow without creating new failure points? What parts of the current operation are stable enough to automate, and what needs standardization first?

KoiGPT decided to answer these questions in the following way. They leverage the latest agentic AI tools to combine continuous diagnostics, reasoning, benchmarking, and forecasting into a real-time digital brain for logistics and manufacturing operations. It allows operations leadership to shift their decisioning from reactive response to proactive optimization, improving overall efficiency and eliminating delays.

But multiple questions often require multiple tools. The grid below illustrates tools that support different aspects of the diagnostic process, from mapping physical flows to stress-testing scenarios and benchmarking maturity. Each tool is most useful when combined with clear questions, defined outputs, and owners who can translate findings into practical changes on the floor (Figure 11).

A common pitfall is that while tools can identify and assess technical opportunities, it's the workforce that determines whether those opportunities become real performance gains. For example, a simulation might show that batching strategy changes would reduce congestion, but the benefits will only be achieved if supervisors understand the new triggers and teams execute the strategy consistently when conditions change.

ADDRESS WORKFORCE AND TALENT CHALLENGES AND IMPACTS

Technology can raise the ceiling, but workforce capabilities ultimately dictate whether an organization operates at that ceiling or below it. In fact, many technology problems are actually talent and workforce problems in disguise—unclear roles, inconsistent routines, uneven training, or reliance on a handful of experts who quietly keep the system up and running with workarounds.

Automation and digital workflows have the potential to dramatically change how work gets done. For example, frontline workers can move from manual execution to system-directed execution supplemented with exception handling. Supervisors can shift from expediting on the floor to managing through signals (e.g., dashboards, queue health, labor balancing). Maintenance and engineering roles can expand into keeping automation running smoothly and increasing the capabilities of controls, not just making mechanical repairs. Inventory control can become more data-driven, with higher expectations for root-cause analysis.

According to this year's survey, leaders are increasingly prioritizing skills related to strategic problem solving (43%) and data analytics (39%) as they report what the most critical supply chain management skills over the next five years will be. The implication for technology deployment is clear: technology roadmaps must be paired with skills roadmaps so new capabilities show up in daily execution as real-world performance gains, not just system features.

Tool/Platform	***Purpose***	***Best Used For***	***Outputs***
Digital Twins	*Virtually replicates a facility/network for scenario testing and optimization*	*Testing "what-if" changes without disrupting operations; comparing alternative designs; stress-testing peak demand*	*Scenario results and trade-offs; bottleneck and congestion hot spots; capacity and throughput projections; ROI inputs for business case*
Warehouse Simulation Software	*Models workflows, capacity, labor, and variability to identify constraints*	*Validating where time is lost (queues, batching, travel); evaluating wave vs waveless, pick methods, shifts; quantifying impact of variability (arrivals, order mix, downtime)*	*Labor and equipment sizing guidance; service-level risk under different demand scenarios*
Warehouse Maturity Models	*Benchmarks capabilities vs best practices across process, technology, data, and people*	*Establishing a baseline and target state; sequencing roadmap initiatives; aligning stakeholders on priorities; comparing sites consistently*	*Maturity score by domain; gap-to-target assessment; prioritized improvement backlog*
AI Opportunity Assessment Frameworks	*Identifies/prioritizes AI and analytics use cases based on value and readiness*	*Selecting high-value, feasible use cases; avoiding "AI for AI's sake"; aligning data readiness, governance, and operating model*	*Use-case short list with value hypotheses; feasibility/readiness scoring*

Figure 11: Diagnostic process for tool selection

IMPROVING COST-TO-SERVE VISIBILITY IN WAREHOUSE OPERATIONS

Situation: One of the biggest challenges in dynamic distribution operations is the ability to understand cost-to-serve variances across products, customers and services. Distributors operating without granular cost visibility often misprice 30–40% of transactions. Systematic correction of this problem through better cost allocation improves EBITDA margins by 1.5–3 percentage points—dramatic gains in an industry where EBITDA margins of 5–8% are typical.

Labor management systems generate a much richer activity data set that, when paired with fully loaded labor rates and automation/equipment cost data, gives operations transaction-level, network-wide visibility into cost-to-serve.

Saddle Creek Logistics Services operates 46 warehousing and fulfillment facilities across the United States in support of hundreds of clients. In large-scale 3PL operations like these, labor is the largest variable expense. The company already had high-level reporting in place, but according to Luke Hendrix, VP of Operations, there was a lack of real-time visibility into how labor hours across multiple clients and sites translated into cost-to-serve. Without this clarity, identifying inefficiencies and calculating actual costs across diverse accounts was a manual, retrospective process.

Action: To gain more granular visibility into its operations, Saddle Creek implemented the Easy Metrics platform at several sites—helping the company assess the performance and business impact of its warehouse operations.

Easy Metrics is a cloud-based warehouse performance management platform that helps companies identify and optimize their labor costs. The company combines its robust technology with a proven cost management methodology to drive 20%+ productivity improvements and 10-15% labor cost savings, while providing significantly greater cost and performance insights into labor-intensive processes. The Easy Metrics platform aggregates the exploding amount of activity data generated by RF scans, RFID, warehouse management systems, ERP systems, time clocks and manual tracking to create detailed cost and performance models.

"From the outset, Saddle Creek's committed focus was on driving measurable improvements across its service delivery model, using data to move beyond high-level reporting and manual analysis," said Steve Cascio, CRO of Easy Metrics. The transformation centered around using activity-based costing to accurately assign labor costs to every workflow and operational touchpoint, replacing retrospective cost analysis with real-time insight. Floor-level, real-time performance dashboards equipped supervisors with the data needed to coach teams and adjust workflows in the moment, while network-level analytics translated complex operational data into actionable recommendations. Throughout the deployment, Saddle Creek leveraged Easy Metrics' best practices and operational expertise to guide the rollout and support change management across participating facilities.

Result: With Easy Metrics' Warehouse Performance Management platform in place, Saddle Creek established a single, trusted view of operational and financial performance. This visibility enabled the organization to identify bottlenecks, eliminate blind spots, and accelerate throughput while improving alignment between operations and finance. As Hendrix noted, "Our collaboration with Easy Metrics has been transformative, moving us beyond simple data collection to a place of true operational intelligence. We're able to make data-driven decisions, resulting in significant efficiency gains and accelerated fulfillment performance for our clients." With greater transparency into cost-to-serve, Saddle Creek expects billing accuracy to improve by clearly connecting labor, workflows, and financial outcomes. As efficiencies continue to be realized, Saddle Creek anticipates labor savings of approximately 3-5%. Beyond near-term gains, Hendrix said the partnership has become "a cornerstone of how we drive value for our clients," reinforcing Saddle Creek's ability to continuously improve service delivery through data-driven decision-making.

TOP 5 SUPPLY CHAIN TRENDS

1) ECONOMIC UNCERTAINTY, INFLATION AND GEOPOLITICAL RISKS

- Outlook is precarious; uncertainty leads to cautious investment strategies and reevaluation of supplier relationships
- Increased costs of goods, services, raw materials, energy, and labor, which is pressuring supply chains and margins
- Trade wars and political instability are disrupting supply chains through tariffs, sanctions, and border closures

2) WORKFORCE, TALENT SHORTAGE, AND CHANGING WORKER SKILLSETS

- Talent shortages cause delays and increased operational costs across supply chains
- Technology is transforming supply chain roles requiring new skills in automation, AI and data analytics
- Industry partnerships help build a skilled talent pipeline for future supply chain workers

3) PACE OF TECHNOLOGY ADOPTION, DIGITIZATION, AND NEED FOR REAL-TIME DATA

- Advanced analytics, AI, IoT, and robotics are disruptors that can enhance efficiency, transparency and visibility
- Adopting new technologies demands investment, use cases, change management, and workforce preparedness
- Early adopters gain competitive advantage

4) SUPPLY CHAIN VISIBILITY, AGILITY, AND RESILIENCY

- Agility enables quick adaptation to market changes and demand shifts while resiliency focuses on recovering from disruptions
- Strong supplier relationships and risk management are key
- AI and IoT empower data analytics and real-time decision-making

5) CYBERSECURITY AND DATA SECURITY

- Digitization is increasing the risk of cyber threats to supply chain operations and sensitive data
- Firms are adopting multi-layer security like encryption, access controls, and continuous monitoring to protect their digital systems
- Educating employees and partners fosters vigilance and resilience against cyber threats

NEXT 5 SUPPLY CHAIN TRENDS

6) RISING COST OF CAPITAL

- Higher capital costs lead companies to delay or scale back infrastructure and tech investments
- Firms are prioritizing high-return projects and seeking alternative funding
- Strong credit profiles and strategic planning help secure favorable terms and sustain growth

7) INVENTORY CHALLENGES

- Pandemic disruptions revealed the need for more responsive and agile inventory management strategies
- Use of AI forecasting tools and supplier collaboration improves inventory accuracy and efficiency
- Achieving optimal inventory requires strategic planning, technology support, and agile execution

8) E-COMMERCE GROWTH

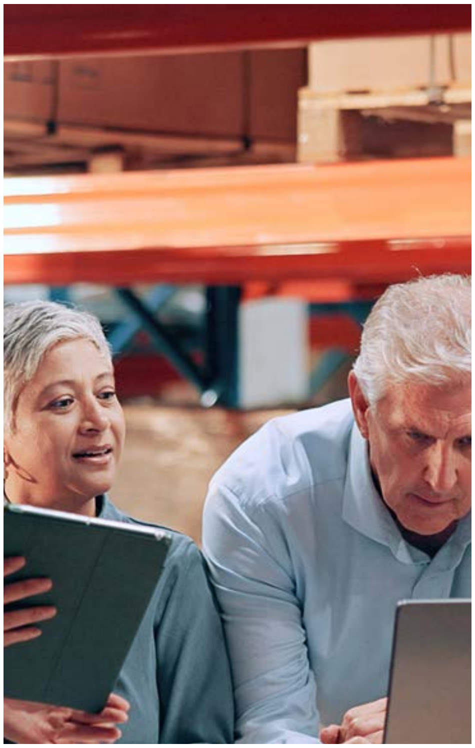
- Same-day or next-day delivery expectations are forcing supply chains to adopt micro-fulfillment and last-mile delivery innovation
- Inventory must be visible and synchronized across stores, warehouses, and online platforms
- Warehouses and DCs are deploying robotic picking systems, automated sorters, and conveyor systems to improve performance

9) CUSTOMER-CENTRICITY

- Designing and operating supply chains with the end customer’s needs, preferences, and experience at the center
- Real-time data, predictive analytics, and AI demand forecasting provide real-time order tracking and multiple delivery channels
- Adoption of AI driven inventory systems can send products to regional hubs before customers order for faster delivery

10) RESHORING

- Accelerating due to tariffs, geopolitical risks, supply chain disruptions, and the need for resilience
- Improves responsiveness and sustainability, while lowering risk of disruption
- Higher operational and labor costs can be offset through automation, robotics, IoT, AI, and advanced analytics



The biggest threat we face isn't disruption—it's the failure to innovate and the risk of running tomorrow's operations on yesterday's equipment and technology.

John Paxton, CEO, MHI

CONCLUSION

In 2026, supply chains are navigating unprecedented volatility—marked by geopolitical tension, climate disruption, cyber threats, and constant market shifts. Yet amid these external pressures, one internal risk overshadows all others: outdated and disconnected technology. While many organizations push to innovate, those still operating on aging infrastructure, manual workflows, or fragmented data systems face dramatically higher operational exposure.

The message of this report is clear: Legacy supply chain equipment and systems are no longer a slow-burning weakness—it is an immediate threat to resilience, competitiveness, and continuity.

Visibility and transparency have become strategic necessities, but legacy systems block progress on critical capabilities such as:

- Real-time demand sensing
- End-to-end inventory visibility
- Advanced analytics and forecasting
- Multi-tier data sharing

Without a unified digital backbone, organizations make decisions from stale or incomplete data—precisely when speed and accuracy are most essential.

AI is no longer optional; it is a foundational requirement for future operations. AI is reshaping forecasting, routing, fulfillment, and decision-making. However, outdated systems create major obstacles such as:

- Data silos undermining model training
- Manual processes slowing automation
- Legacy ERPs failing to integrate with modern analytics
- Rigid architectures blocking scalability

Companies that fail to modernize will fall behind as AI-enabled competitors achieve lower costs, faster cycle times, and greater agility.

Geopolitical and economic turbulence further elevates risk. Constant shifts in trade policy, tariffs, and regional instability require supply chains to reroute logistics, reconfigure suppliers, and redesign networks in real time. Legacy systems—built for a more predictable era—cannot:

- Model scenarios
- Evaluate new suppliers
- Adjust transportation flows
- Support nearshoring or reshoring at speed

In today's environment, agility is survival—and outdated technology locks organizations into slow, brittle processes.

Progress begins with clear operational assessments, coordinated digital and on-the-floor improvements, and disciplined performance metrics. These foundations enable companies to evolve from optimizing isolated functions to rewiring the entire supply chain. In this next stage, planning and execution operate as a single connected system, not a chain of disconnected handoffs. AI and advanced analytics function as a "digital nervous system", continuously translating signals into decisions, decisions into execution, and execution into learning.

This transformation is complex. It depends on modern platforms, data harmonization, strong integration, and consistent governance. This is not a one-time project; it is a continuous innovation cycle where organizations steadily enrich data, refine decision logic, and expand AI autonomy.

Physical AI and automation extend this rewiring into the operational world—but the true competitive advantage will come from the company-wide structure that runs it: standards, controls, and a workforce equipped to operate, improve, and scale automated systems.

Workforce evolution is equally urgent. Leading supply chain organizations are closing the digital skills gap, while those clinging to outdated tools struggle to attract and retain talent in analytics, automation, and AI. New generations expect intuitive interfaces, automation support, AI-driven insights, and cloud-native systems. Legacy platforms don't just slow operations—they weaken an organization's ability to compete for the talent required to run a modern supply chain.

Today's supply chains must be:

- Real-time
- Interconnected
- Predictive
- Cloud-based and agile
- Integrated with risk management

Without innovation, companies can face deeper blind spots, slower responses, and increasing vulnerability to financial and operational shocks.

Organizations that integrate AI-driven orchestration and physical automation into one cohesive operating model will not just endure disruption—they will turn it into momentum, energizing their workforce and rewiring operational complexity into sustained performance and growth.

Recognize that transformation is an ongoing cycle; continuously enhance data, refine decision-making, and expand AI autonomy to adapt to evolving conditions. Equip your workforce with the necessary skills to leverage new technologies, investing in training to ensure teams can effectively manage and scale automated solutions. By reframing disruption as an opportunity, leaders who integrate operational efficiency with AI orchestration and physical automation will turn challenges into growth, maintaining a competitive advantage for years to come.

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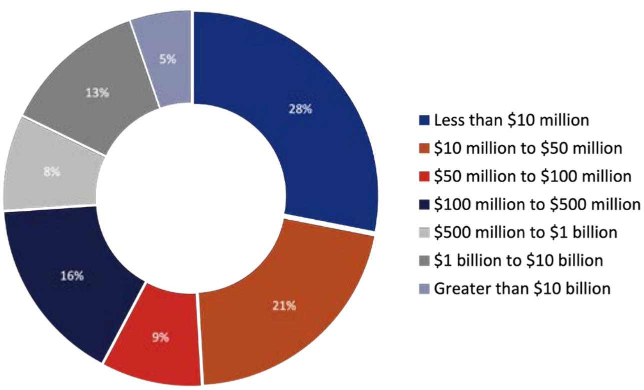
ABOUT THE REPORT

The 2026 MHI Annual Industry Report is our thirteenth annual study of emerging disruptive technologies and innovations that are transforming supply chains around the world. The findings are primarily based on an in-depth global survey conducted in late 2025, which involved over 500 supply chain professionals from a wide range of company types and industries.

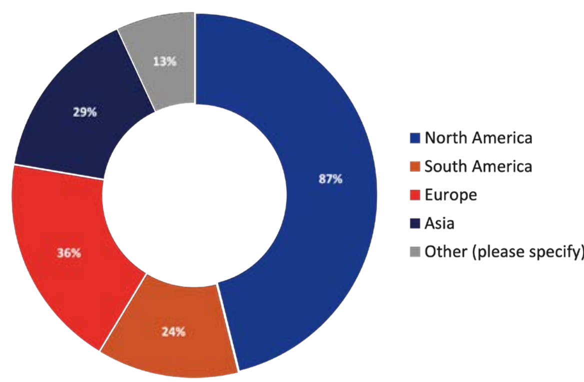
60% of the participants are executives with the role of CEO, Vice President, General Manager, Manager, or Department Head. Participating companies range in size from small to large, with 75% reporting annual sales in excess of $50 million, and 18% reporting annual sales of $1 billion or more.

COMPANY SIZE BY REVENUE (USD)

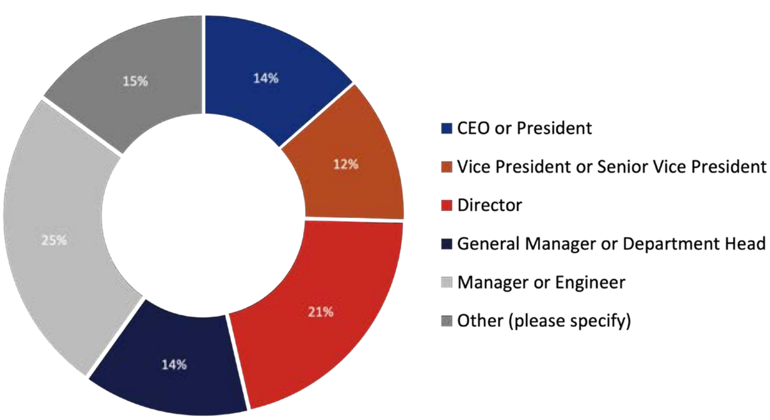


RESPONDER PROFILE BY GEOGRAPHY

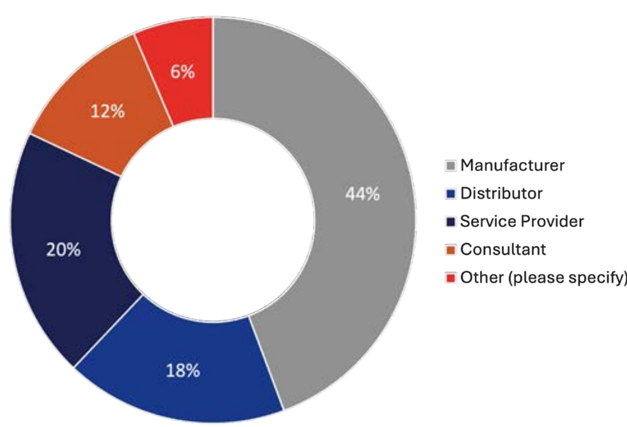
(SELECT ALL THAT APPLY)



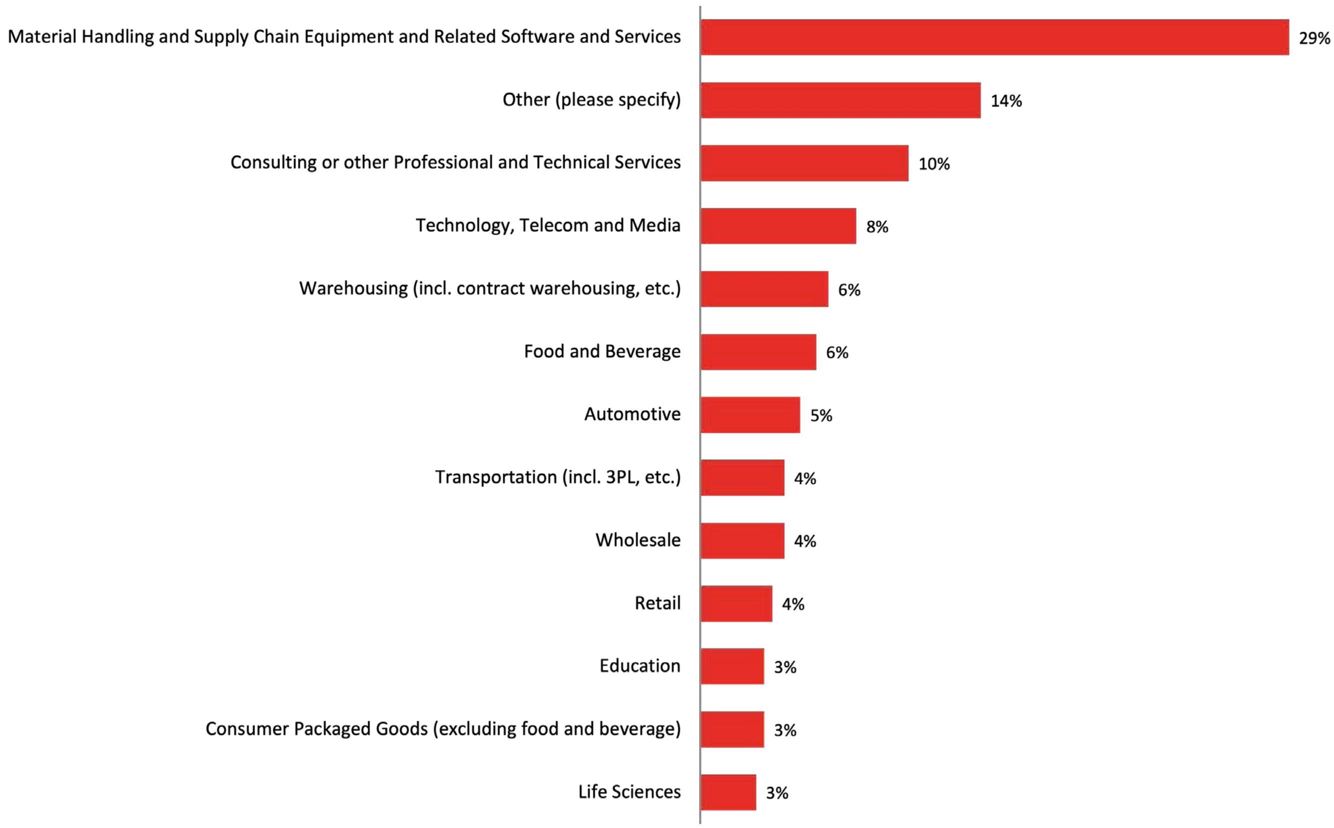
RESPONDER'S ROLE



RESPONDER'S COMPANY TYPE



RESPONDER'S INDUSTRY



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We would like to acknowledge the hundreds of organizations that participated in our survey. We would also like to thank the MHI Board for their contributions to the survey and conclusions.

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